NM4QM 2026 · HANDS-ON SESSION · 45 MIN

# Spin and anomalous Hall effects with WannierBerri and GPAW

Symmetry-adapted Wannier functions, spin–orbit coupling on demand, the spin Hall effect of graphene and the anomalous Hall effect of the altermagnet MnTe



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DIPC, Donostia-San Sebastián · 13–16 October 2026

THE STORY

# Two puzzles, two acts

| 0–8 min | 8–20 | 20–37 | 37–45 |
| --- | --- | --- | --- |
| Setup and ideas | Act 1: graphene | Act 2: MnTe | Your material, questions |

## TWO PUZZLES

Graphene: spin–orbit coupling opens a gap, a quantum spin Hall insulator

MnTe: no net magnetisation, yet an anomalous Hall effect

## WHAT YOU WILL DO

Wannier functions from GPAW, SOC on demand, Hall conductivities

SETUP

# Before we start: one check

```
cd wannier-berri-demo && conda activate wbdemo
python setup/check_setup.py        # should end with "Ready."
cd 01_graphene
python 1_dft.py                    # 10 s
jupyter lab graphene.ipynb
```

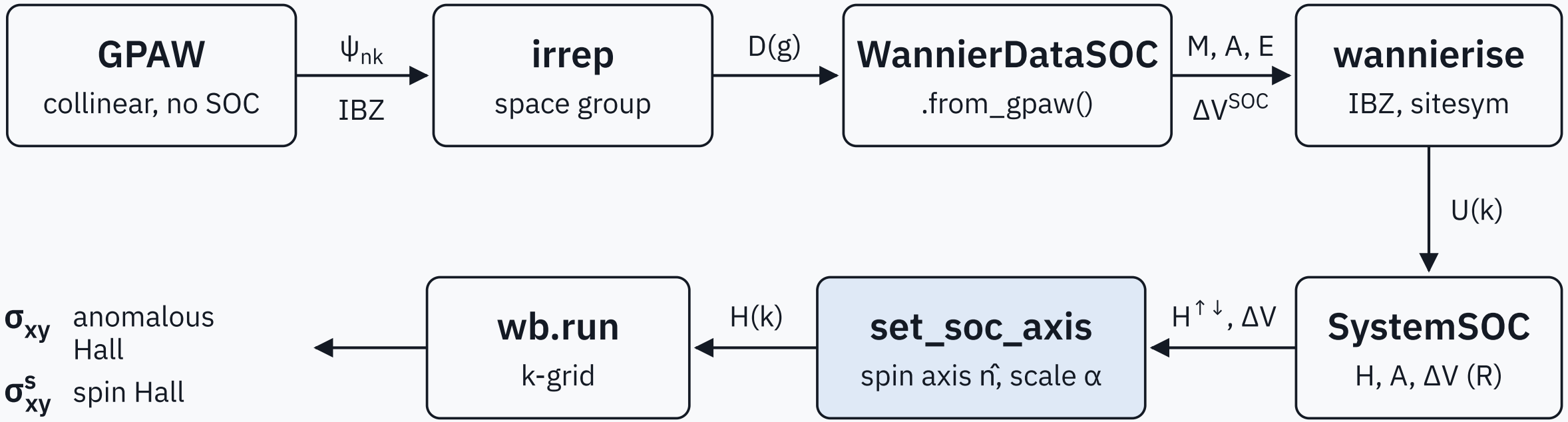
No environment: `bash setup/setup.sh`

Fell behind: `bash run_all.sh`

DFT fails: `tar -xzf dft-files.tar.gz`

Numbers on slides: checkpoints

# One Python pipeline, GPAW to conductivities




Each box is one call; the shaded step needs no new DFT and no new Wannierisation

# Berry curvature needs millions of k-points; DFT gives tens

Intrinsic anomalous Hall conductivity:

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int_{\mathrm{BZ}} \frac{d^3k}{(2\pi)^3} \sum_n f(\varepsilon_{n\mathbf{k}})\, \Omega_n^z(\mathbf{k})$$

$$\Omega_n^z = -2\,\mathrm{Im} \sum_{m \neq n} \frac{v_{nm}^x\, v_{mn}^y}{(\varepsilon_n - \varepsilon_m)^2}$$

$f$: occupation, $\varepsilon_n$: band energy, $v_{nm}^a = \langle u_n | \partial_{k_a} H | u_m \rangle$. $\Omega_n^z$ is large and narrow wherever two bands come close.

Wannier functions give $H(\mathbf{k})$ at any $\mathbf{k}$:

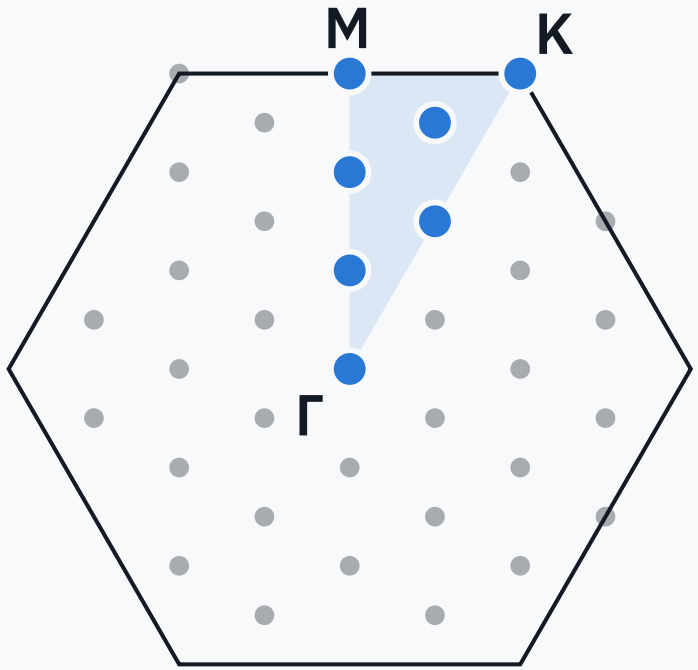
$$H_{ij}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}}\, \langle w_{i\mathbf{0}} | \hat{H} | w_{j\mathbf{R}} \rangle$$

$w_{j\mathbf{R}}$: Wannier function $j$ in the cell at $\mathbf{R}$; a few hundred $\mathbf{R}$ are enough.

# Symmetry-adapted Wannier functions: DFT on irreducible k only

Each Wannier function transforms like its orbital

Degeneracies come out exact




6 × 6 grid: 36 points, 7 irreducible (blue)

METHODS

# Wannierisation: WannierBerri vs Wannier90

$$\Omega = \sum_n \left[ \langle w_n | r^2 | w_n \rangle - \left| \langle w_n | \mathbf{r} | w_n \rangle \right|^2 \right] = \Omega_{\mathrm{I}} + \tilde{\Omega}$$

$\Omega$: spread · $\Omega_{\mathrm{I}}$: fixed by the subspace (disentanglement) · $\tilde{\Omega}$: set by the gauge $U(\mathbf{k})$ (localisation)

| | **Wannier90** | **WannierBerri, this demo** |
|---|---|---|
| Input | .amn, .mmn, .eig files from a DFT interface | computed in Python from the GPAW wave functions |
| k-points | full grid; irreducible ones with site_symmetry | irreducible ones |
| Algorithm | disentangle first, then minimise $\tilde{\Omega}$ by steepest descent or conjugate gradients | one loop: a disentanglement step and a local gauge update per iteration, no gradient descent |
| Symmetry | site_symmetry with a .dmn file from pw2wannier90, not with a frozen window | from irrep, for any code it reads; frozen window allowed |
| SOC | spinor DFT run with SOC | collinear run, SOC added afterwards; spin ↓ from symmetry in altermagnets |

Marzari, Vanderbilt, PRB 56, 12847 (1997) · Souza, Marzari, Vanderbilt, PRB 65, 035109 (2001) · Sakuma, PRB 87, 235109 (2013) · Pizzi et al., JPCM 32, 165902 (2020)

# Post-processing: WannierBerri vs postw90

$$H(\mathbf{K}+\boldsymbol{\kappa}) = \sum_{\mathbf{R}} e^{i\boldsymbol{\kappa}\cdot\mathbf{R}} \left[ e^{i\mathbf{K}\cdot\mathbf{R}}\, H(\mathbf{R}) \right]$$

$\boldsymbol{\kappa}$: points of a coarse FFT grid · $\mathbf{K}$: shifts of that grid; one FFT per shift, the shifts together give the dense grid

| | **postw90** | **WannierBerri** |
|---|---|---|
| $H(\mathbf{R}) \to H(\mathbf{k})$ | sum over **R** at every k, cost $\propto N_{\mathbf{R}}$ | FFT over sub-grids, cost $\propto \log N_{\mathrm{FFT}}$ per k |
| Symmetry | full grid | irreducible k-points only, result symmetrised |
| Hot spots | finer mesh where \|Ω\| exceeds a threshold | recursive adaptive refinement |
| Fermi levels | cost grows with their number | many at almost no extra cost |
| SOC axis | fixed by the DFT run | `set_soc_axis` : any axis and strength |

Up to three orders of magnitude faster on dense grids

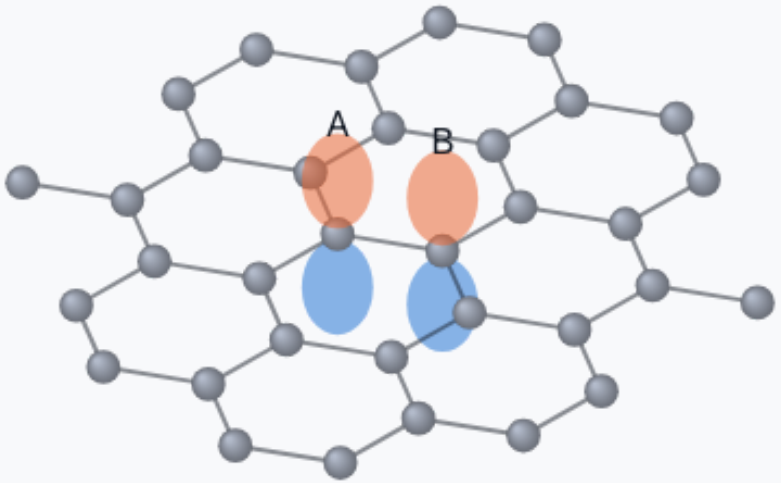
Wang, Yates, Souza, Vanderbilt, PRB 74, 195118 (2006) · Tsirkin, npj Comput. Mater. 7, 33 (2021) · Pizzi et al., JPCM 32, 165902 (2020)

# Does graphene have a gap?

| Step | Run | Checkpoint |
|---|---|---|
| 0 · DFT | `python 1_dft.py` | `gpaw/nscf.gpw` |
| 1 · Symmetry | irrep | 48 operations, 7 of 36 k-points |
| 2 · Wannier functions | `wannierise` | two equal $p_z$-like functions |
| 3 · Bands | interpolation | blue on grey in the window |
| 4 · SOC on demand | `set_soc_axis` | gap ≈ 36 μeV (GPAW ≈ 39 μeV) |
| 5 · Spin Hall | `wb.run` | plateau at e/2π |

Kane, Mele, PRL 95, 226801 (2005)

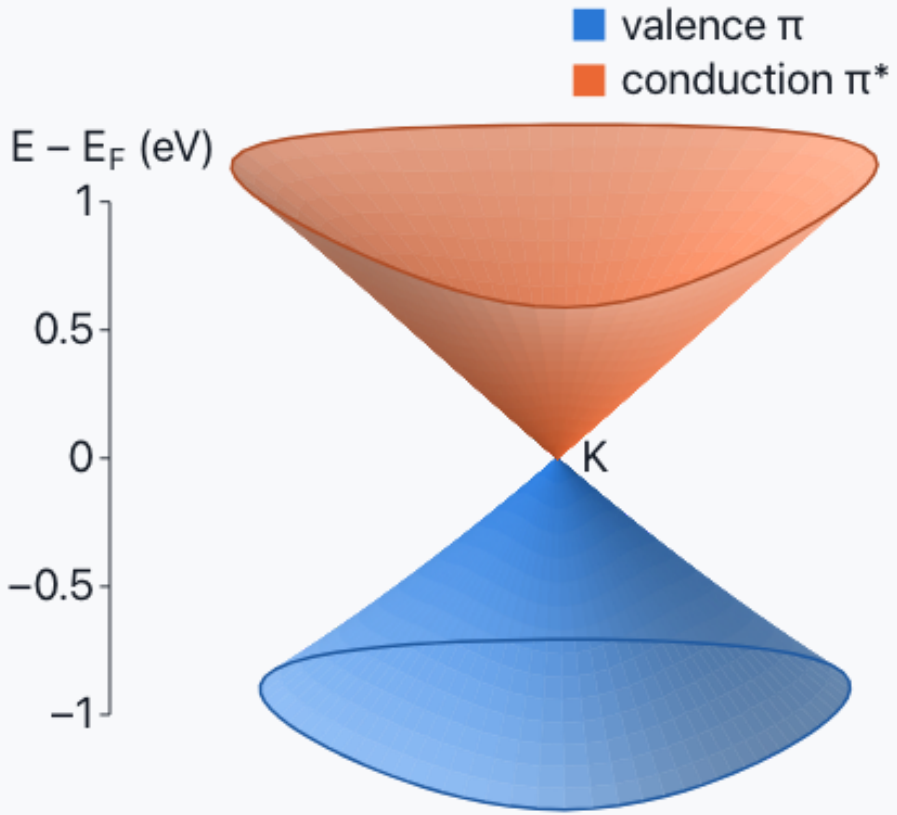
# Graphene: two $p_z$ Wannier functions from GPAW



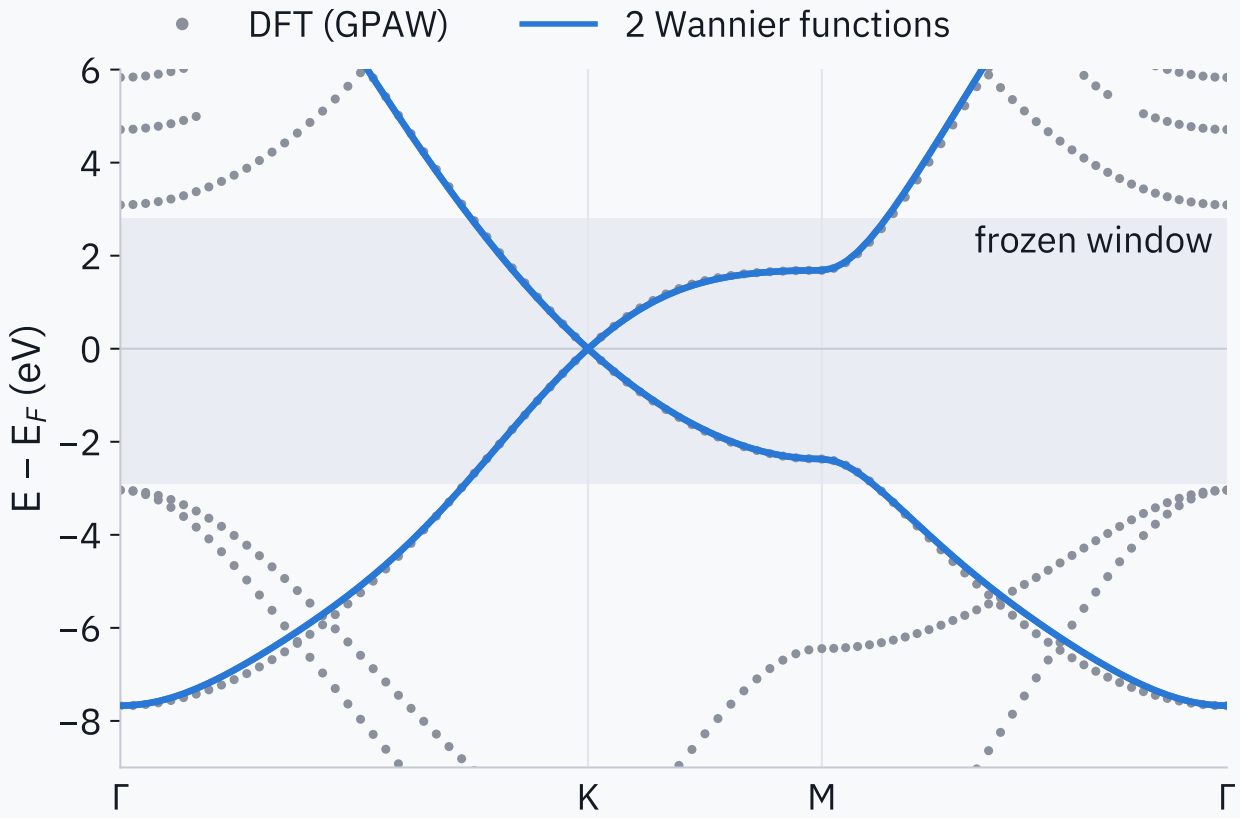

One $p_z$ Wannier function on each carbon, A and B

```
calc = GPAW(mode=PW(400), xc="PBE",
            kpts={"size": (6, 6, 1), "gamma": True})
atoms.calc = calc
atoms.get_potential_energy()
sg = SpaceGroup.from_gpaw(calc)                    # irrep
kpts = sg.get_irreducible_kpoints_grid((6, 6, 1))  # 7 of 36
calc.fixed_density(kpts=kpts, nbands=20).write(
    "gpaw/nscf.gpw", mode="all")

# graphene.ipynb
pz = Projection(position_num=[[1/3, 2/3, 0], [2/3, 1/3, 0]],
                orbital="pz", spacegroup=sg)
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=ProjectionsSet([pz]))
wandata.wannierise(froz_min=EF - 2.9, froz_max=EF + 2.8,
                   frozen_states={0: [1]},  # pi at Gamma
                   num_iter=100, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata, berry=True)
```




The Dirac cone of the Wannier model at K



# SOC as a matrix you can rotate and scale

$$H(\mathbf{k}) = H_0(\mathbf{k}) \otimes \sigma_0 + \alpha \sum_c \Delta V^c(\mathbf{k}) \otimes \sigma_c(\hat{\mathbf{n}})$$

$H_0$: no SOC · $\sigma_c(\hat{\mathbf{n}})$: Pauli matrices along $\hat{\mathbf{n}}$ · $\alpha$: SOC scale

$\Delta V$: the PAW matrix elements of GPAW's SOC

New $\hat{\mathbf{n}}$ or $\alpha$: no new DFT

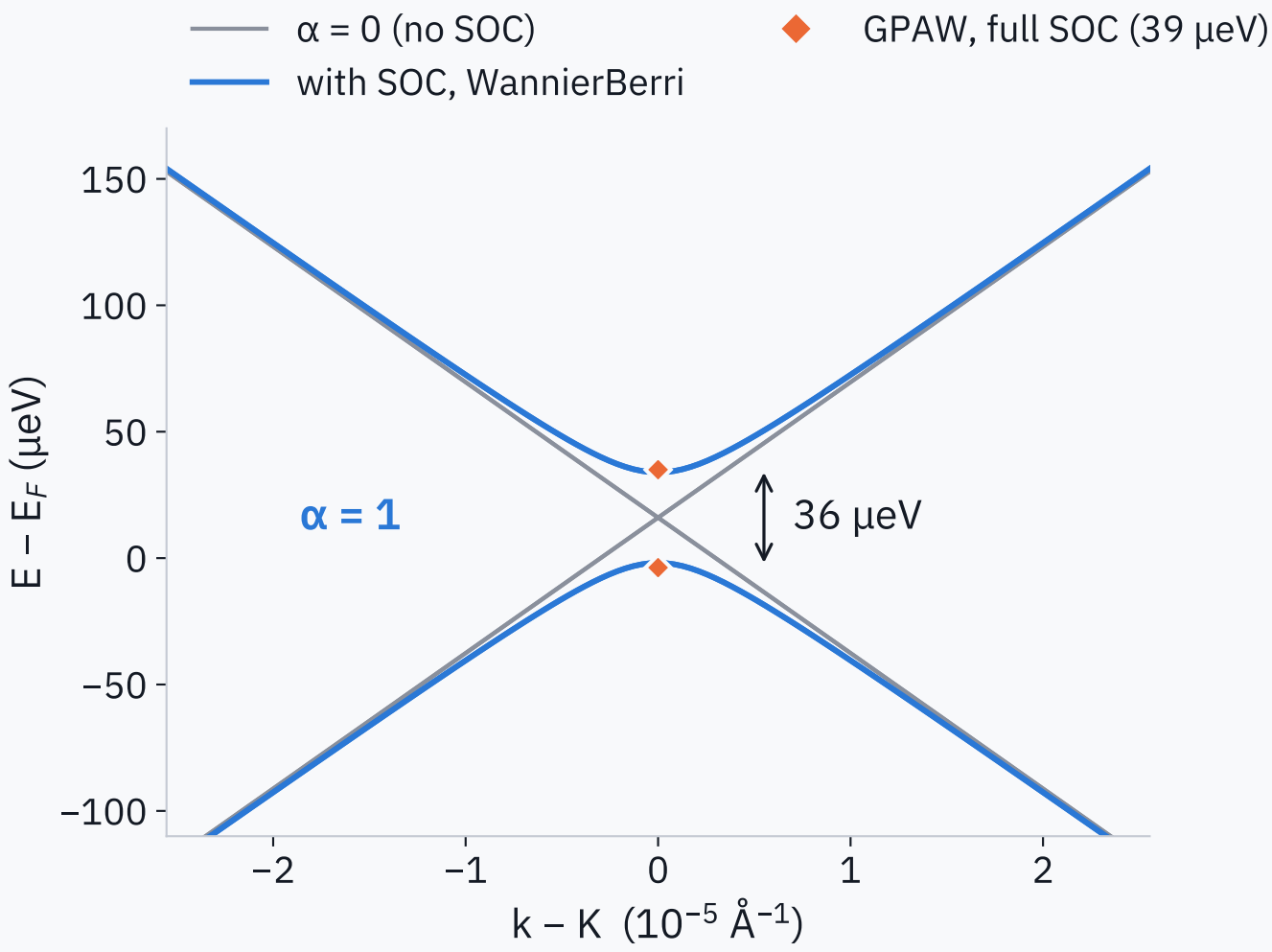
```
K = [1/3, 1/3, 0]
energy = wb.calculators.tabulate.Energy(
    degen_thresh=1e-8)
for alpha in [0, 1]:
    system.set_soc_axis(alpha_soc=alpha)
    E = wb.evaluate_k(system, k=K,
            calculators={"E": energy})
```

```
              gap at K
alpha = 0      0
alpha = 1     36 ueV
GPAW SOC      39 ueV
```

## SOC gap at the Dirac point K

| | |
|---|---|
| Wannier model, two $p_z$ | **36 µeV** |
| GPAW, all 20 bands | **39 µeV** |
| Experiment (Sichau 2019) | **42 µeV** |

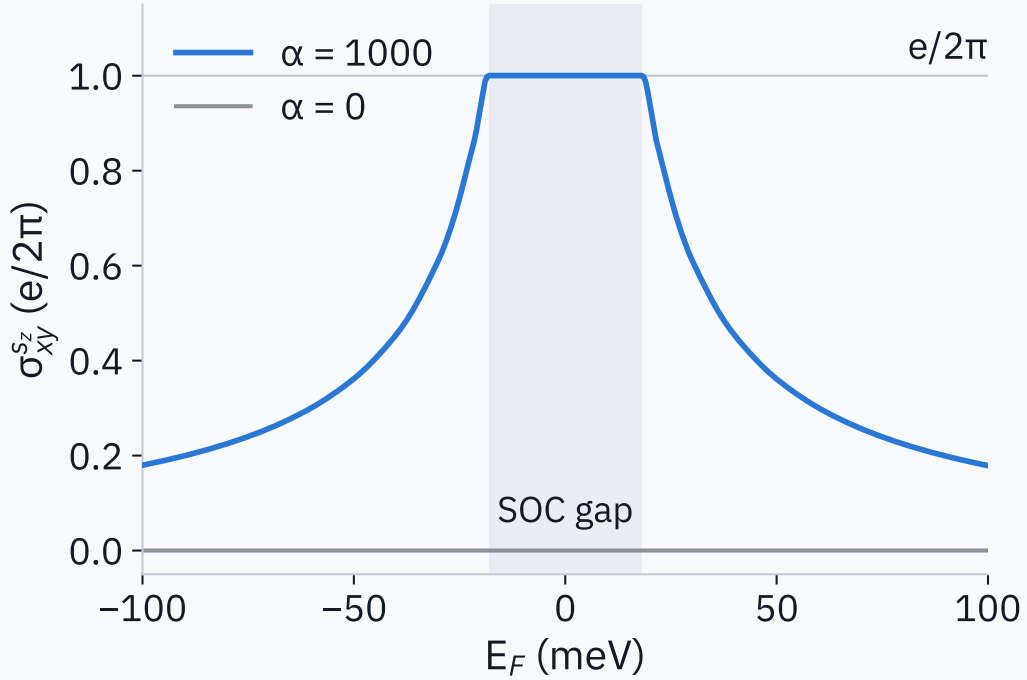
From the d part of $p_z$ (Gmitra 2009)



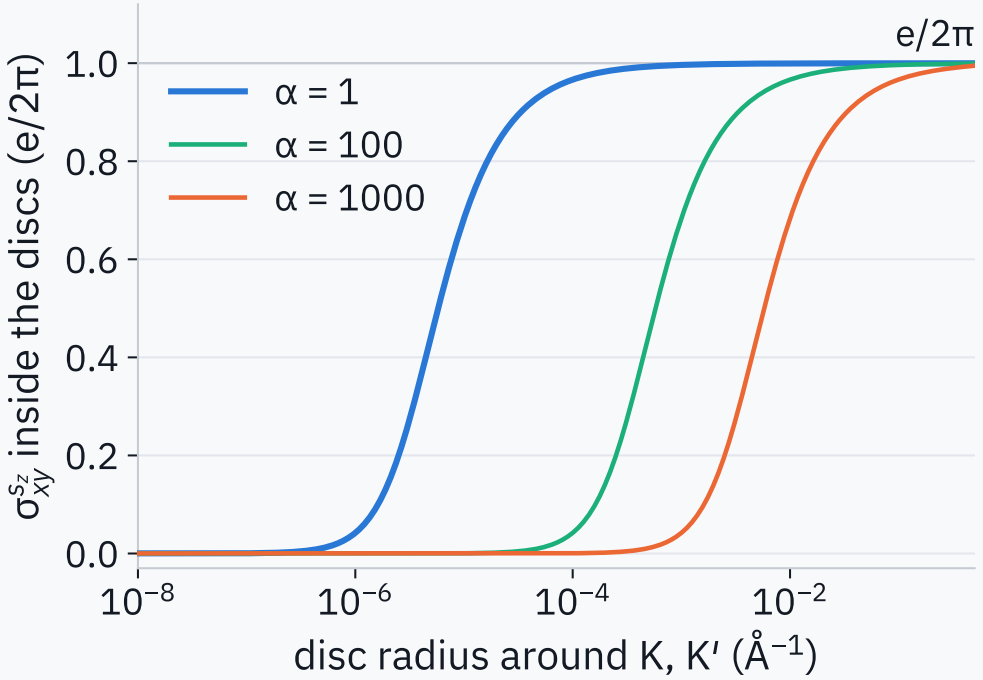

Kane, Mele, PRL 95, 226801 (2005) · Gmitra et al., PRB 80, 235431 (2009) · Sichau et al., PRL 122, 046403 (2019)

$$\sigma_{xy}^{s_z} = \frac{e}{\hbar}\int \frac{d^2k}{(2\pi)^2}\sum_n f_{n\mathbf{k}}\,\Omega_n^{s_z}, \qquad \Omega_n^{s_z} = -2\hbar^2\,\mathrm{Im}\sum_{m\neq n}\frac{(j_x^{s_z})_{nm}(v_y)_{mn}}{(\varepsilon_n-\varepsilon_m)^2}$$

$j_x^{s_z} = \frac{1}{2}\{s_z, v_x\}$: spin current, $v$: velocity. With $s_z$ conserved, $\sigma_{xy}^{s_z} = (e/2\pi)(C_\uparrow - C_\downarrow)/2 = e/2\pi$




$\alpha$ = 1000 (gap 36 meV): plateau at e/2π




$\alpha$ = 1, the real 36 μeV gap: also e/2π

ACT 2 · MNTE.IPYNB

# A Hall effect without magnetisation

| Step | Run | Checkpoint |
|---|---|---|
| 0 · DFT | `python 1_dft.py` | moments ≈ ±4.4 $\mu_B$ |
| 1–2 · Spin ↑ Wannier | `altermagnetic=True` | 24 of 144 k-points |
| 3 · Splitting | no SOC | up to ≈ 1 eV at $k_z \neq 0$ |
| 4 · Turn $\hat{\mathbf{n}}$ | `set_soc_axis` | SOC shift at A 4–5× smaller in plane |
| 5 · Anomalous Hall | three $\hat{\mathbf{n}}$ | 0 for x and c, up to ≈ 80 S/cm for y |
| 6 · Rotate in plane | 0–30° | far from sin 3φ |

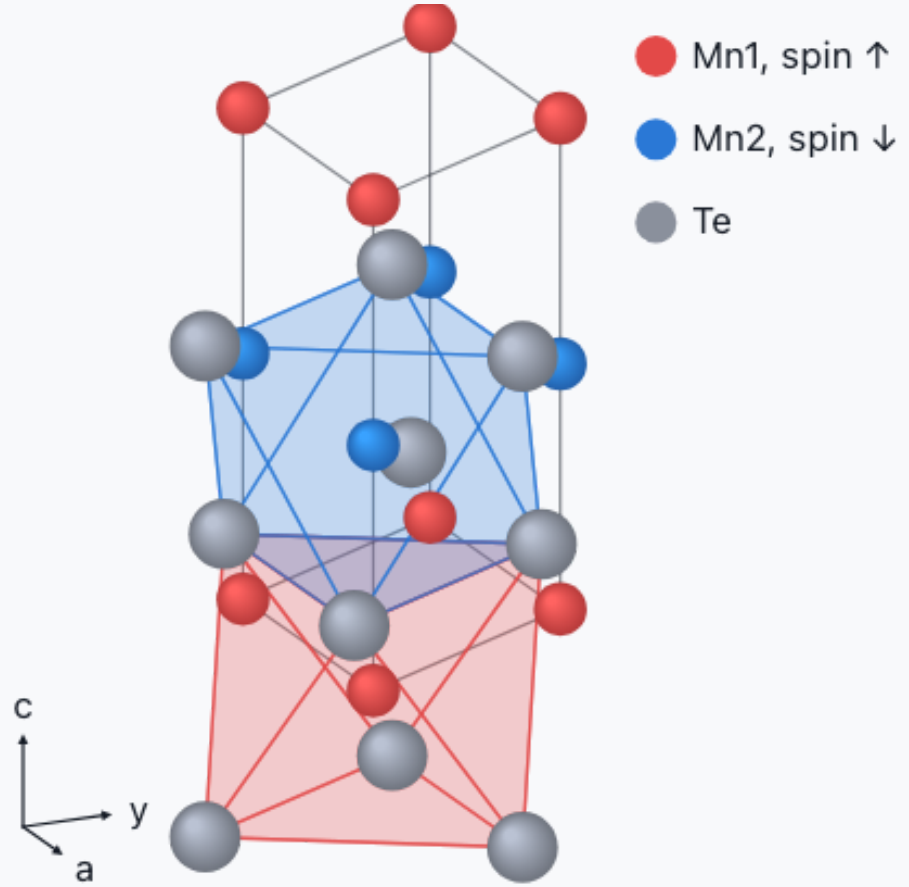
Gonzalez Betancourt et al., PRL 130, 036702 (2023)

# Altermagnets: zero net moment, spin-split bands without SOC

Spin ↑ and ↓ sublattices, related by a rotation

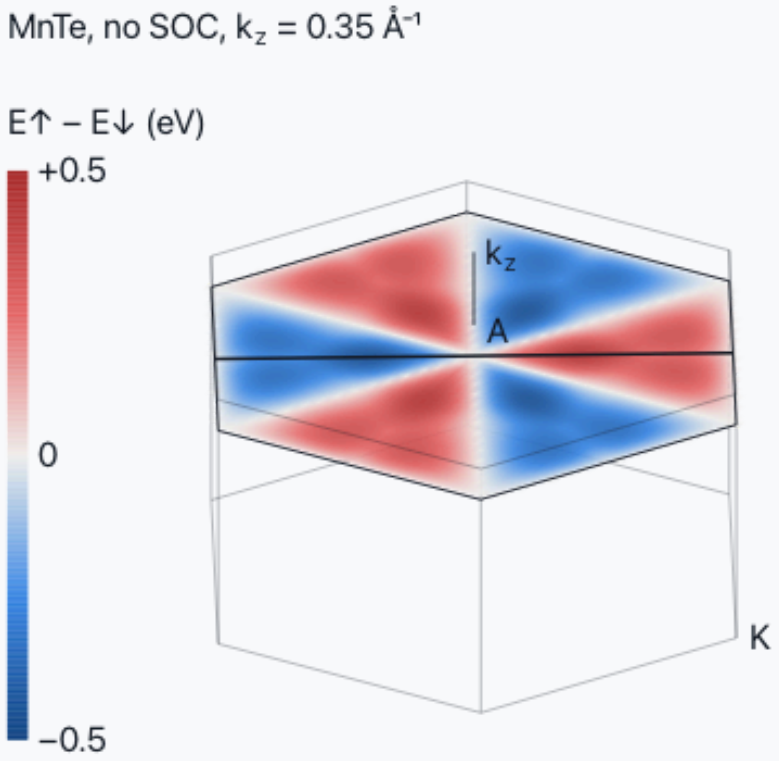
Zero net moment, yet an anomalous Hall effect

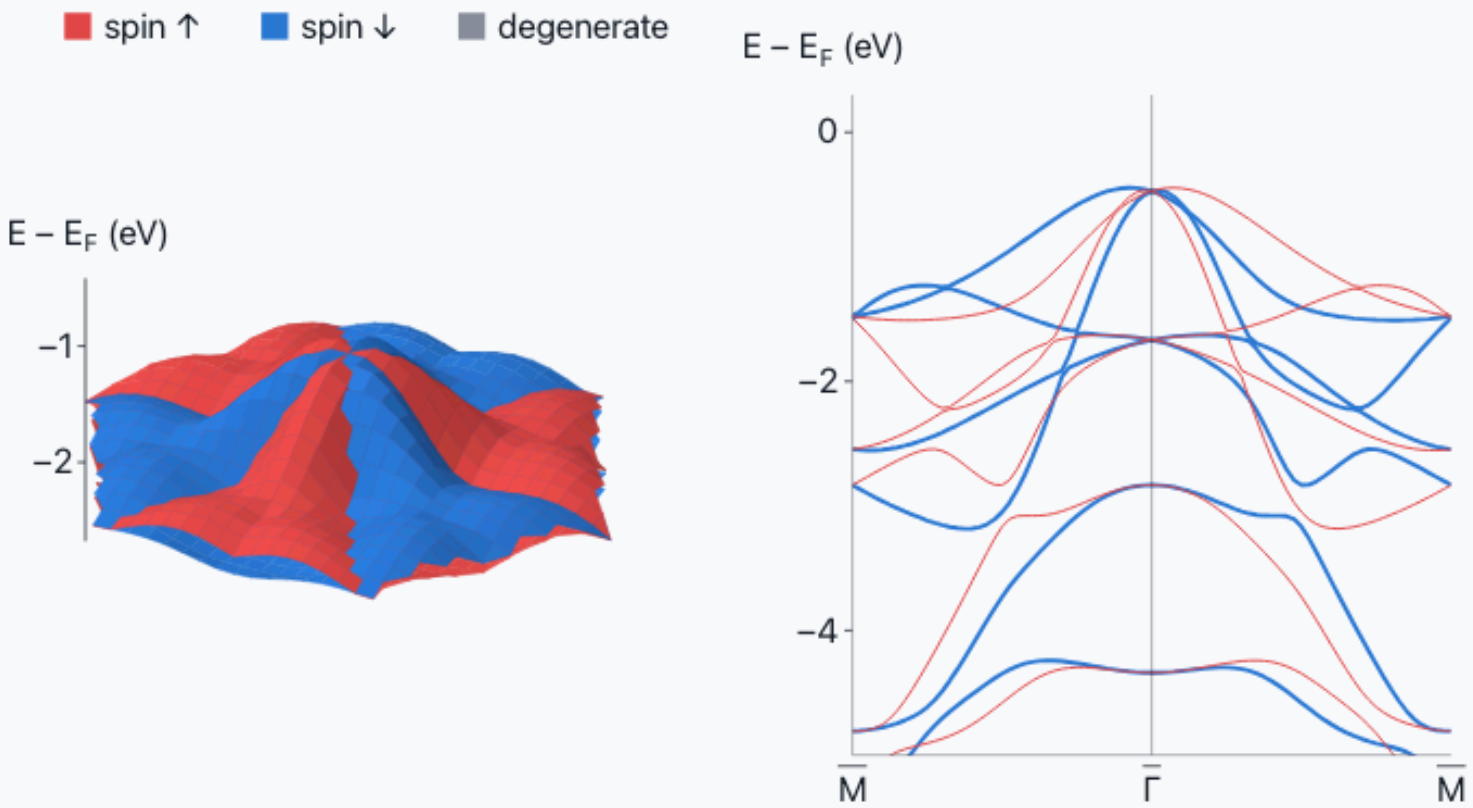
**MnTe**: altermagnetic up to $T_N \approx 310$ K



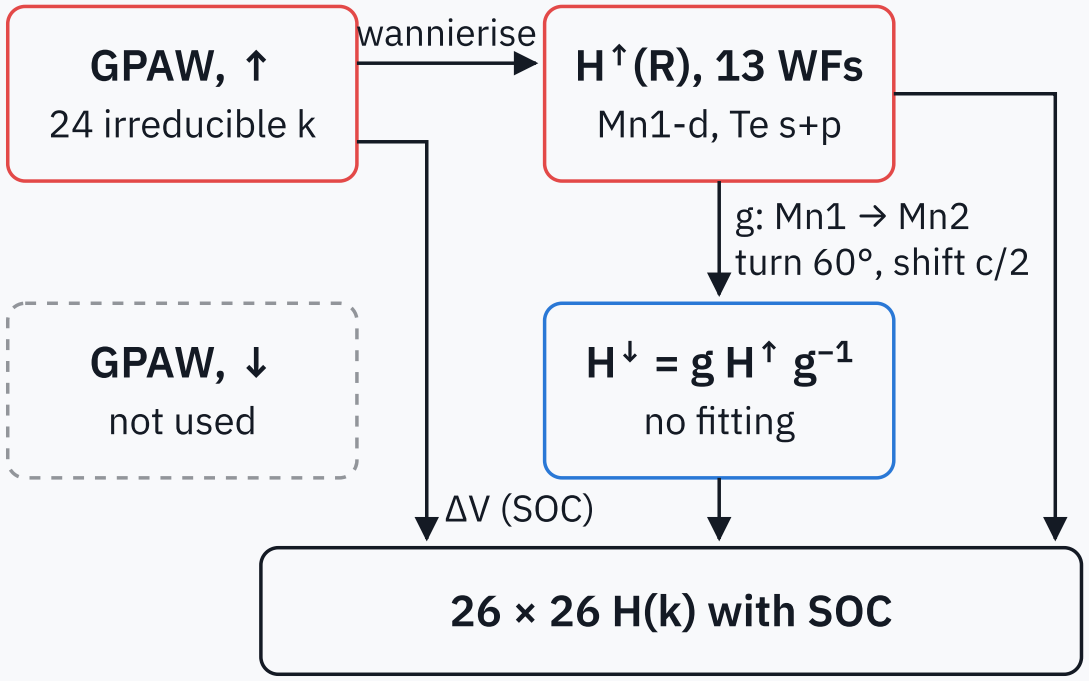

Te octahedra of Mn1 (↑) and Mn2 (↓), turned by 60°

# Spin ↓ bands are spin ↑ bands at a rotated **k**: the splitting changes sign, g-wave



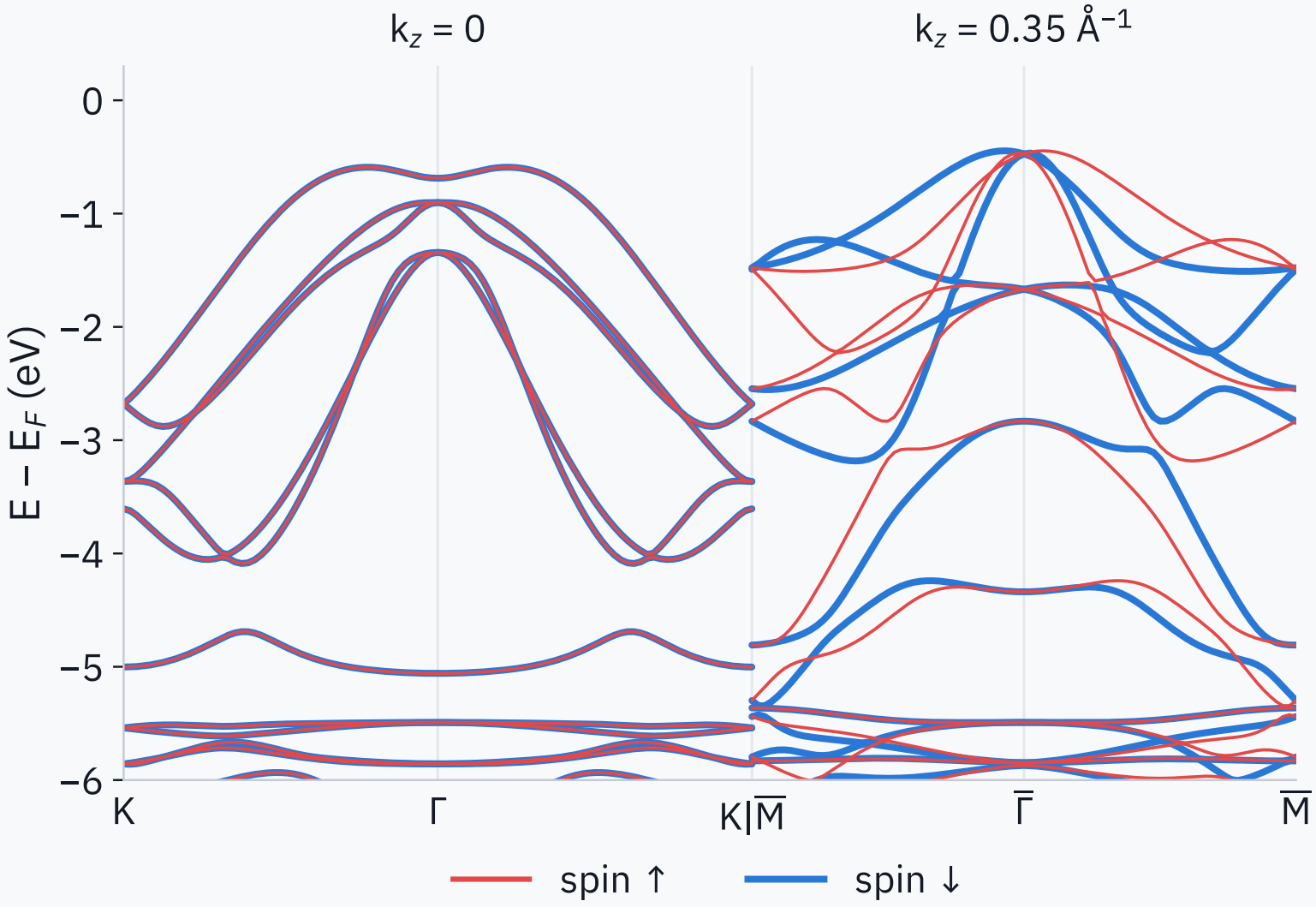



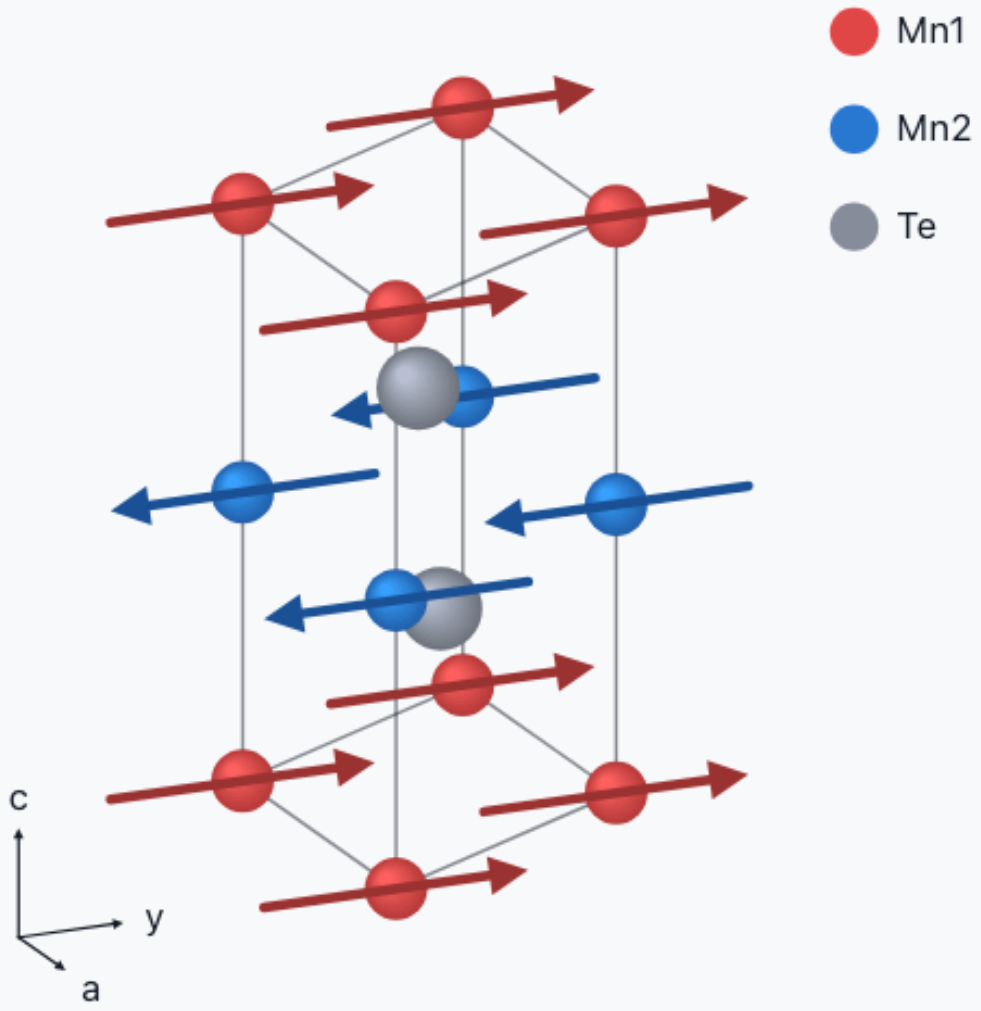
# Wannierise spin ↑ only, spin ↓ follows by symmetry

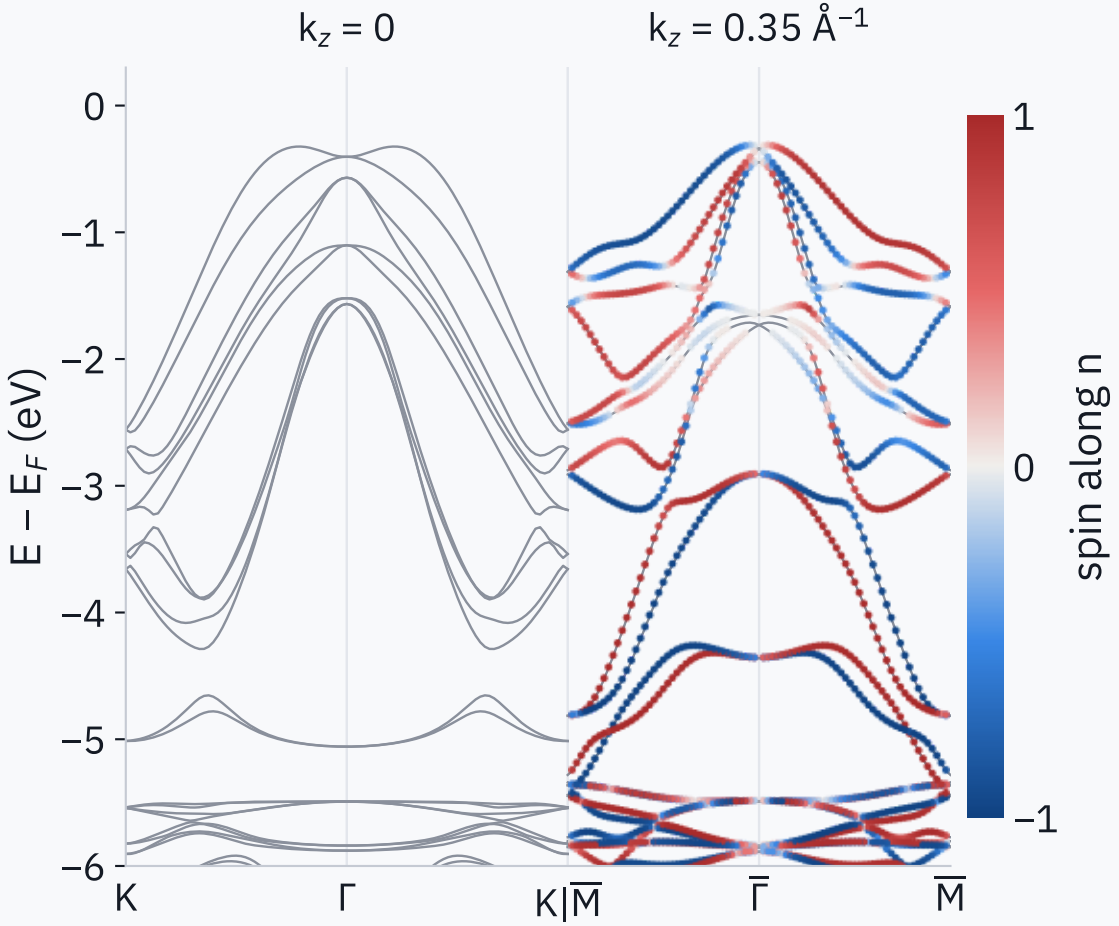



```
calc = GPAW("gpaw/nscf.gpw")   # PBE+U, 24 of 144 k
sg = SpaceGroup.from_gpaw(calc)
Te = [[1/3, 2/3, 1/4], [2/3, 1/3, 3/4]]
projections = ProjectionsSet([
    Projection(position_num=[0, 0, 0], orbital="d",
               spacegroup=sg),          # Mn1 only
    Projection(position_num=Te, orbital="sp2",
               spacegroup=sg, xaxis=[0, -1, 0],
               rotate_basis=True),
    Projection(position_num=Te, orbital="pz",
               spacegroup=sg)])
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=projections,
    altermagnetic=True,   # spin down from symmetry
    IBstart=8)            # skip Mn 3s, 3p
wandata.wannierise(froz_min=-np.inf,
                   froz_max=EF + 0.1,
                   num_iter=200, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata,
                                       berry=True)
```

Dudarev et al., PRB 57, 1505 (1998) · Kriegner et al., PRB 96, 214418 (2017) · Bossini et al., NJP 22, 083029 (2020)


$k_z = 0$
$k_z = 0.35$ Å$^{-1}$
0
−1
−2
−3
−4
−5
−6
$E - E_F$ (eV)
K
Γ
K|M
Γ
M
spin ↑
spin ↓

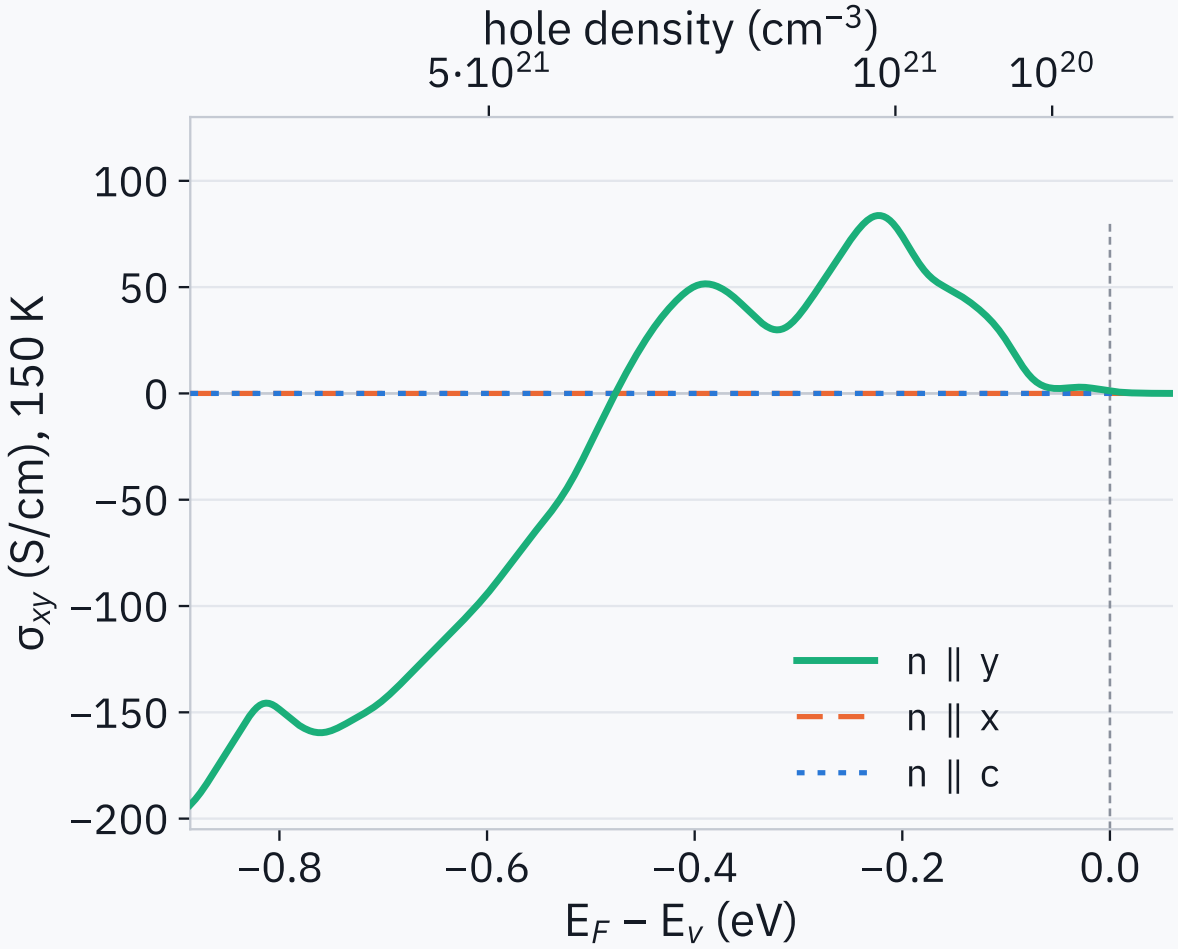



$\hat{\mathbf{n}} \parallel$ y, colour: spin along $\hat{\mathbf{n}}$

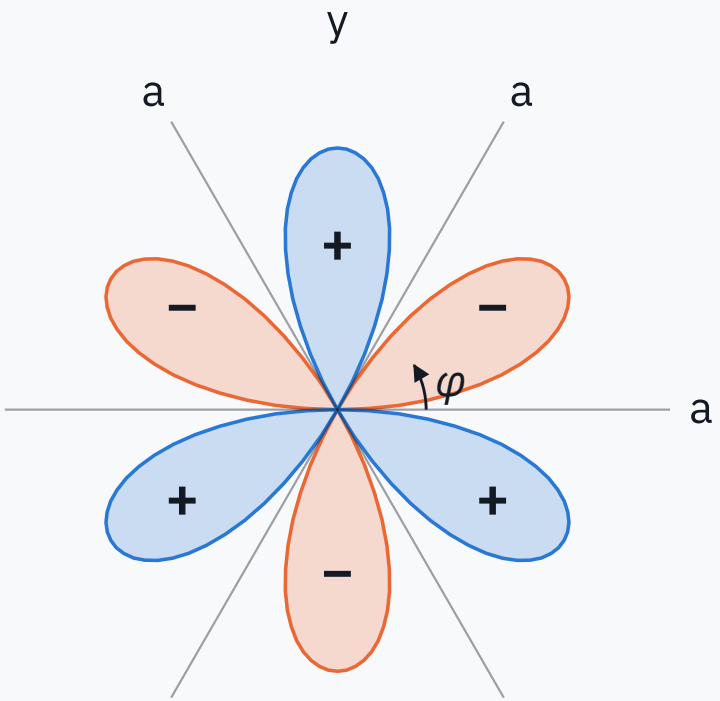
```
ahc = wb.calculators.static.AHC(Efermi=Efermi,
    degen_thresh=1e-4, smoother=smoother)
directions = {"x": (90, 0), "y": (90, 90),
              "z": (0, 0)}   # theta, phi
grids = {"x": (48, 48, 32), "y": (96, 96, 64),
         "z": (48, 48, 32)}
for name, (theta, phi) in directions.items():
    system.set_soc_axis(theta=theta, phi=phi,
                        units="degrees")
    result = wb.run(system,
        grid=wb.Grid(system, NK=grids[name]),
        calculators={"ahc": ahc})
```

Zero for $\hat{\mathbf{n}} \parallel x$ and $\hat{\mathbf{n}} \parallel c$, by symmetry

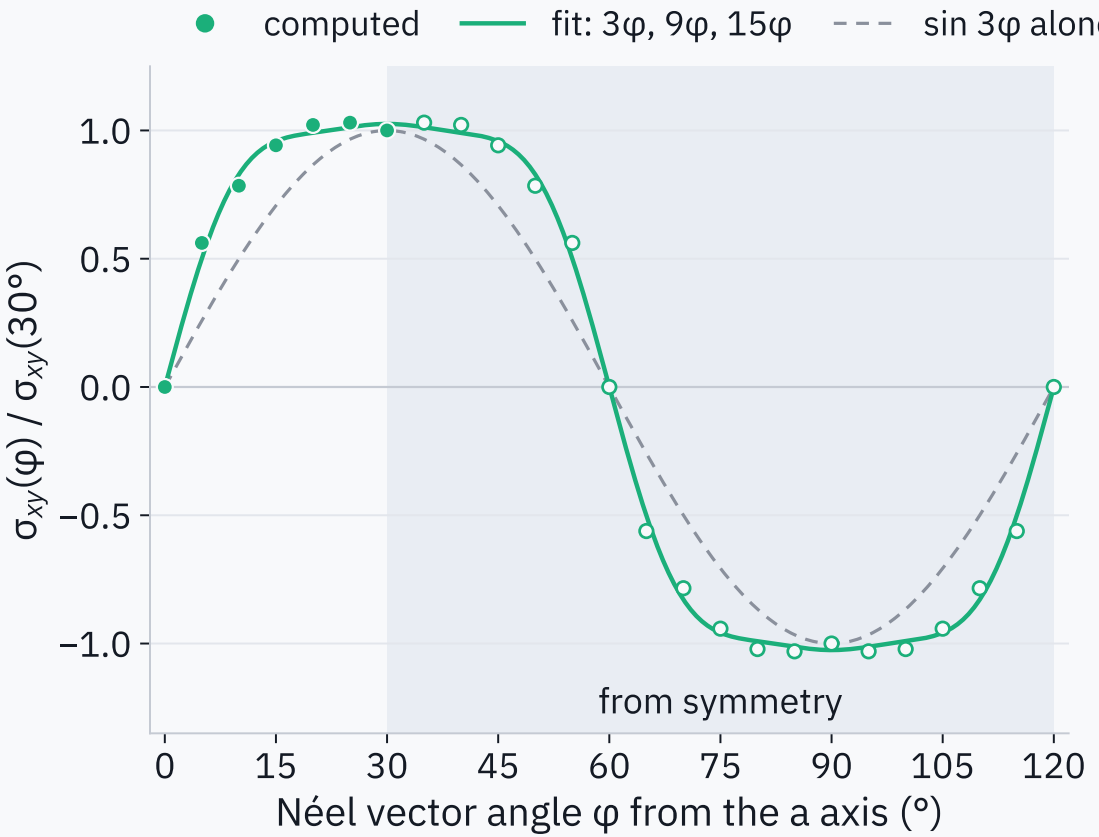


In-plane Néel vector at angle φ from a, amplitudes $A_m$ in S/cm:

$$\sigma_{xy}(\varphi) = A_3 \sin 3\varphi + A_9 \sin 9\varphi + A_{15} \sin 15\varphi + \ldots$$




$\sigma_{xy}$ = 0 along every a axis, sign flip every 60°




Filled: computed. Open: from symmetry

# Your material next

```
calc = GPAW(...)                                  # SCF
kpts = SpaceGroup.from_gpaw(calc)\
           .get_irreducible_kpoints_grid(grid)
calc.fixed_density(kpts=kpts, nbands=...)  # NSCF
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=...)
wandata.wannierise(froz_min=..., froz_max=...,
                   sitesym=True)
system = wb.SystemSOC.from_wannierdata(
    wandata, berry=True)
system.set_soc_axis(theta, phi)
wb.run(system, grid=..., calculators=...)
```

Project on the orbitals of your bands

Freeze every state you need

Check a few levels against GPAW first

SUMMARY

# Take-aways, and how to run it

Symmetry: DFT only on irreducible k-points

SOC on demand: any spin axis, no new DFT

Graphene: spin Hall plateau e/2π

Altermagnets: Wannierise one spin, get both

**RUN IT**

github.com/yaroslav-zhumagulov/wannier-berri-demo

```
bash setup/setup.sh    # once, 5 min
conda activate wbdemo
bash run_all.sh        # 6 min, or the notebooks
```

Linux, macOS, WSL2 · GPAW 26.7 from conda-forge

BACKUP

# Details

Kane–Mele form · nodal planes of MnTe · timings · references

# The Wannier SOC matrix has the Kane–Mele form

$$H_{\mathrm{SO}} = i\lambda \sum_{\langle\langle ij \rangle\rangle} \nu_{ij}\, c_i^{\dagger} s_z c_j, \qquad \Delta_{\mathrm{SO}} = 6\sqrt{3}\,\lambda$$

- Second neighbours: **$\lambda \approx 3.4$ μeV**, purely imaginary, sign ±1 as in Kane–Mele
- $6\sqrt{3}\,\lambda \approx 36$ μeV: the whole gap of the full model
- $\Delta V^{x,y} = 0$: the mirror plane forbids Rashba terms, $s_z$ is conserved
- No SOC between first neighbours: each bond lies in a vertical mirror

| Shell | $\lvert d \rvert$ (Å) | $\lvert t \rvert$ (eV) | $\lvert \Delta V^{x,y} \rvert$ (eV) | $\lvert \Delta V^{z} \rvert$ (μeV) |
|---|---|---|---|---|
| 1st, A–B | 1.42 | 2.90 | 0 | 0 |
| 2nd, A–A | 2.46 | 0.25 | 0 | **3.4** |
| 3rd, A–B | 2.84 | 0.26 | 0 | 0 |
| 4th, A–B | 3.76 | 0.02 | 0 | 0 |
| 5th, A–A | 4.26 | 0.04 | 0 | 0 |
| 6th, A–A | 4.92 | 0.03 | 0 | 0.3 |

Largest element per shell of $H(\mathbf{R})$ and $\Delta V(\mathbf{R})$; nothing fitted

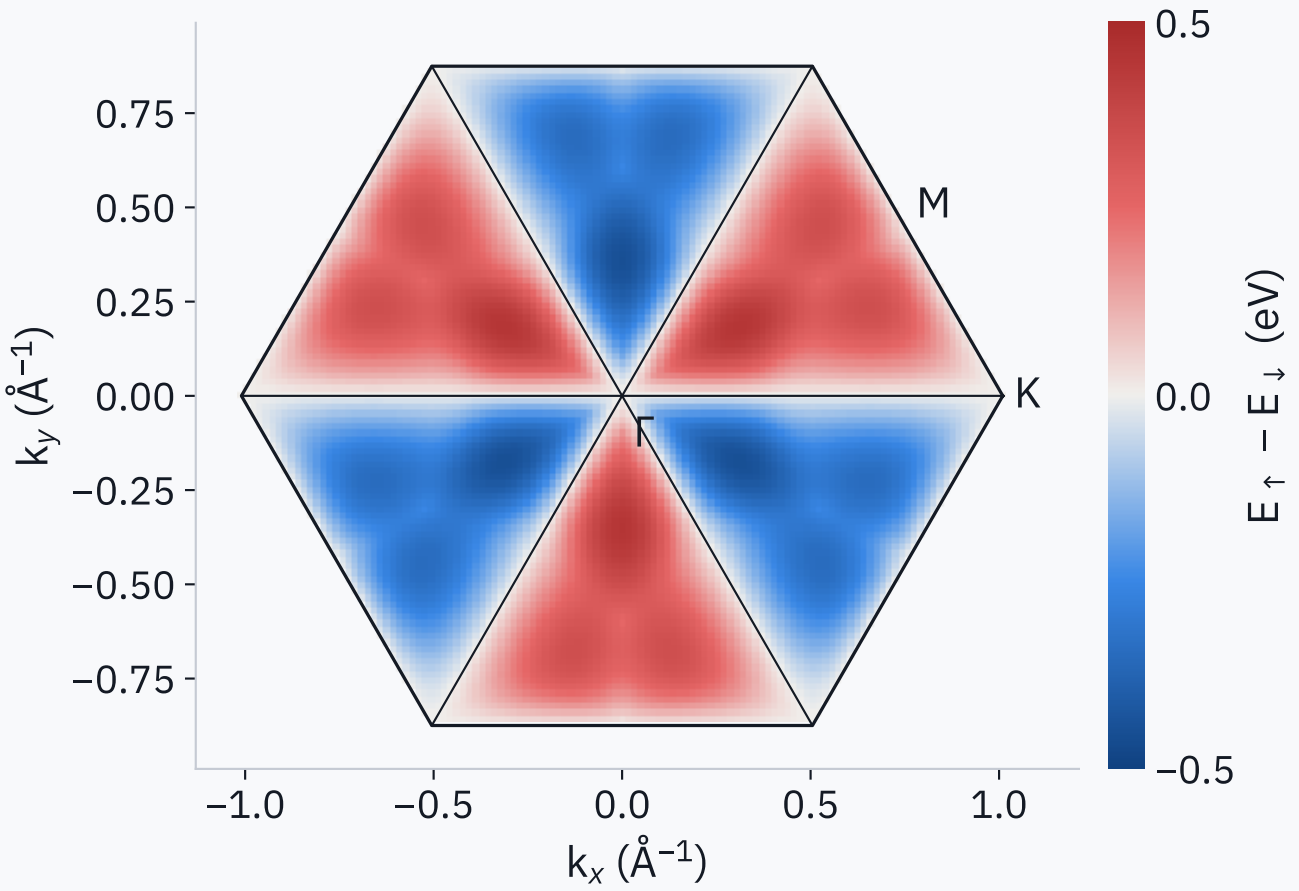
Kane, Mele, PRL 95, 226801 (2005)

# Four nodal planes make MnTe g-wave

$E_\uparrow(\mathbf{k}) = E_\downarrow(R_g\mathbf{k})$ for a spin-flipping $g$, so the splitting vanishes on planes that $R_g$ leaves in place:

- $m_z$: $k_z = 0$ and $k_z = \pi/c$
- three $c$-glides: three vertical planes through Γ–K

In the model the splitting is exactly zero on these planes, and up to ≈ 1 eV along Γ–M at $k_z = 0.35$ Å$^{-1}$




$E_\uparrow - E_\downarrow$, top valence band, $k_z = 0.35$ Å$^{-1}$

BACKUP

# Where the time goes

| Step | Graphene | MnTe |
|---|---|---|
| GPAW SCF + NSCF, one core | 10 s | 45 s |
| `WannierDataSOC.from_gpaw` | 3.3 s | 9.7 s |
| `wannierise` | 2.1 s | 9.7 s |
| `SystemSOC.from_wannierdata` | 1.0 s | 6.6 s |
| Conductivities (step 5) | 28 s | 63 s |
| Néel scan (step 6) | – | 66 s |
| Animation frames | 1 s | 22 s |
| Whole notebook | 1.1 min | 4.2 min |

Apple M2, 8 cores, 8 Ray workers: 6 minutes for everything, try-its and bonus steps included

With the Wannier model in hand, a new spin axis, SOC strength or Fermi level costs seconds

# References


## CODES

- WannierBerri: S. S. Tsirkin, npj Comput. Mater. **7**, 33 (2021)
- GPAW: J. J. Mortensen et al., J. Chem. Phys. **160**, 092503 (2024)
- IrRep: M. Iraola et al., Comput. Phys. Commun. **272**, 108226 (2022)
- Wannier90: G. Pizzi et al., J. Phys.: Condens. Matter **32**, 165902 (2020)
- Wannier ecosystem: A. Marrazzo et al., Rev. Mod. Phys. **96**, 045008 (2024)

## METHODS

- N. Marzari, D. Vanderbilt, PRB **56**, 12847 (1997); I. Souza et al., PRB **65**, 035109 (2001); N. Marzari et al., Rev. Mod. Phys. **84**, 1419 (2012)
- R. Sakuma, PRB **87**, 235109 (2013)
- D. Xiao, M.-C. Chang, Q. Niu, Rev. Mod. Phys. **82**, 1959 (2010)
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## PAW, SOC AND DFT+U

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